

SAI BHARGAV THUMMALAPALLI

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Professional Summary

Software Developer / Integration Specialist with **5+ years** of **cumulative hands-on experience** through professional and university-based projects. Skilled in **Java, Spring Boot, REST APIs, Microservices, AWS, Database Integration, Docker, and CI/CD**. Experienced in building **scalable backend services, secure API integrations, and cloud-based solutions**. Strong understanding of **microservices, event-driven architecture, distributed systems, and enterprise integration**.

Technical Skills

- **Programming Languages:** Java, JavaScript, SQL, Bash Scripting
- **Backend Technologies:** Spring Boot, Spring MVC, Hibernate ORM, JDBC, Servlets, JSP, REST APIs, Microservices
- **Integration Technologies:** Enterprise API Integration, REST Integration, System-to-System Integration, API Development, Asynchronous Processing, Event-Driven Architecture Concepts, Distributed Systems
- **API & Security:** RESTful API Design, API Authentication, Authorization, HMAC-SHA256, Secure API Communication, Request Validation, Exception Handling
- **Cloud Platforms:** AWS EC2, AWS RDS, Amazon S3, Azure Fundamentals
- **AWS:** EC2, RDS, S3, IAM Concepts, Cloud Deployment, Cloud Monitoring Concepts
- **Messaging & Event-Driven Architecture:** Apache Kafka, RabbitMQ, Event-Driven Architecture (EDA), Asynchronous Communication, Message Queuing, Pub/Sub, Distributed Systems
- **Databases:** MySQL, PostgreSQL, Redis
- **DevOps & Infrastructure:** Docker, Kubernetes, Jenkins, Terraform, Git, CI/CD Pipelines
- **Testing & API Tools:** JUnit, Mockito, Postman, Bruno
- **Monitoring & Logging:** ELK Stack, Grafana Dashboards, Linux Monitoring Tools, Application Logs, Production Troubleshooting
- **Software Engineering:** OOP, SOLID Principles, Design Patterns, SDLC, Agile, Code Reviews, Clean Code, Problem Solving
- **AI Tools:** Gemini, Claude, MCP Servers, GPTs

Work Experience

Opensols Infotech Pvt Ltd — Software Developer

Jun 2023 – Aug 2026

- Developed and maintained enterprise-grade backend and integration services using **Java, Spring Boot, REST APIs, and Microservices Architecture**.
- Designed and implemented RESTful APIs enabling secure communication between internal application components and external systems.
- Developed reusable integration components following principles of **loose coupling, modularity, scalability, and maintainability**.
- Implemented API request validation, authentication mechanisms, authorization, exception handling, and secure communication.
- Worked on integration workflows involving backend services, databases, external APIs, and business applications.
- Applied microservices architecture principles to create modular and independently maintainable backend services.
- Worked with relational and NoSQL databases including **MySQL, PostgreSQL, and MongoDB**.
- Optimized SQL queries, database operations, API processing, and backend workflows to improve application performance.
- Assisted with cloud deployment and environment configuration using **AWS, Docker, and CI/CD pipelines**.
- Used **Git, Jenkins, Docker, and Terraform** to support automated build, deployment, and infrastructure workflows.
- Monitored application logs and system health using Linux tools and logging frameworks to identify and resolve integration and production issues.
- Troubleshoot production issues by analyzing API requests, application logs, database operations, and service behavior.
- Collaborated with frontend developers, QA engineers, project managers, and technical stakeholders to deliver end-to-end solutions.
- Participated in requirements analysis, solution design, development, testing, deployment, maintenance, and production support.
- Participated in Agile ceremonies including sprint planning, daily stand-ups, code reviews, retrospectives, and technical discussions.
- Developed unit tests using **JUnit and Mockito** to improve application reliability and code quality.
- Applied software design patterns including **Factory Pattern and Builder Pattern** to build extensible and maintainable application components.

Projects

Core Payment Integration System

2025

- Designed and developed a secure **payment integration platform** connecting merchant applications with multiple payment providers.

- Built modular **Spring Boot microservices** for payment processing, transaction management, validation, and provider integration.
- Designed REST APIs for communication between merchant systems, internal services, and external payment providers.
- Implemented secure request authentication using **HMAC-SHA256**.
- Applied integration patterns to maintain separation between payment providers and core business logic.
- Implemented the **Factory Design Pattern** to support multiple payment-provider implementations while reducing system coupling.
- Implemented transaction persistence using **Spring JDBC and MySQL**.
- Optimized database queries and transaction-processing workflows for improved performance.
- Implemented request validation, exception handling, logging, and failure-handling mechanisms.
- Developed unit tests using **JUnit and Mockito** to ensure service reliability.
- Deployed backend services on **AWS EC2** and configured database infrastructure on **AWS RDS**.

University-Based Software Development & Integration Projects

2019 – 2023

- Gained extensive hands-on software development experience through multiple **university-based software engineering and integration projects**.
- Designed and developed backend applications using **Java, Spring Boot, REST APIs, databases, and microservices concepts**.
- Built applications involving API communication, system integration, database persistence, authentication, and business workflow processing.
- Developed RESTful services and integrated frontend applications with backend APIs.
- Applied **Object-Oriented Programming, SOLID principles, design patterns, and modular architecture** to project implementations.
- Worked with relational and NoSQL databases including **MySQL, PostgreSQL, and MongoDB**.
- Performed application testing, debugging, performance optimization, and deployment activities.
- Used Git for version control and followed collaborative software development practices.
- Gained practical exposure to **cloud computing, Docker, CI/CD, distributed systems, and DevOps concepts**.
- Participated in requirements analysis, technical design, development, testing, documentation, and project demonstrations.

Cloud-Based CI/CD Pipeline Automation

2025

- Built an automated CI/CD pipeline for continuous integration, automated testing, and application deployment.
- Integrated **Git, Jenkins, Docker, and Terraform** to streamline software delivery workflows.
- Containerized backend applications using Docker for consistent deployment across environments.
- Implemented infrastructure provisioning and deployment automation using **Terraform**.
- Reduced manual deployment effort and improved deployment consistency through automation.
- Applied DevOps practices to improve release efficiency, deployment reliability, and application availability.

Education

Lovely Professional University — Bachelor of Technology in Computer Science Engineering

2019 – 2023

- Graduated with a CGPA of **7.28**.
- Completed practical software engineering projects involving backend development, API integration, databases, cloud technologies, and DevOps.
- Studied core subjects including Data Structures, Algorithms, Operating Systems, Database Management Systems, Computer Networks, and Software Engineering.
- Applied academic concepts to practical application development, system integration, and software engineering projects.

Sri Chaitanya Junior College — Intermediate (MPC)

2017 – 2019

- Completed Mathematics, Physics, and Chemistry stream.
- Developed strong foundations in analytical thinking, mathematics, and logical problem solving.

Greenwood High School — SSC

2016 – 2017

- Completed Secondary School Certificate (SSC) with **8.8 CGPA**.

Certifications

- **Core Java Made Easy — Udemy**
 - Advanced Java programming, OOP, collections, exception handling, multithreading, and file handling.
- **JDBC, Servlets and JSP — Java Web Development Fundamentals — Udemy**
 - Java web development using Servlets, JSP, JDBC, MVC architecture, and database connectivity.
- **Spring Boot 3, Spring 6 & Hibernate for Beginners — Udemy**
 - Spring Boot, Spring MVC, Hibernate ORM, REST APIs, dependency injection, and enterprise backend development.
- **The Ultimate MySQL Bootcamp — Udemy**
 - SQL, joins, indexing, normalization, database design, and query optimization.
- **Linux Command Line Bootcamp — Udemy**
 - Linux command-line operations, shell scripting, process management, permissions, and troubleshooting.